HEMAL MEWANTHA MUNASINGHE

Data Scientist | AI/ML Engineer | Agentic AI Specialist

+94 76 543 6732 | mewanmuna2000@gmail.com | Kelaniya, Sri Lanka | LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

Results-driven Data Science undergraduate (GPA 3.82/4.00) with hands-on experience in Machine Learning, Large Language Models, and Agentic AI systems. Proven ability to architect end-to-end AI pipelines — from data ingestion and vector embeddings to RAG-based retrieval and multi-agent orchestration. Passionate about building production-ready AI solutions that deliver measurable, real-world impact.

EXPERIENCE

Data Science Intern

Oct 2025 – Apr 2026

International Water Management Institute (IWMI) — Pelawatte, Sri Lanka

- Contributed to data science initiatives at a leading international research organization focused on water resource management.
- Worked with real-world datasets and applied statistical methodologies to support research objectives.

UNDERGRADUATE RESEARCH

Multimodal Emotion Recognition System

Core Technologies: Deep Learning · Multimodal AI · Active Learning (UBAL) Model Quantization (Edge AI) · Computer Vision · Audio Processing · Grad-CAM

- Developed a dual-branch neural network combining facial features (MTCNN) and audio features (Log-Mel Spectrograms) for emotion recognition on a Sinhala video dataset.
- Integrated Grad-CAM interpretability to analyze and visualize visual vs. audio feature contributions, improving model transparency and prediction understanding.
- Investigated model quantization strategies for Edge AI deployment, targeting inference optimization on resource-constrained devices.

EDUCATION

B.Sc. (Hons) in Data Science

2021 – Present

Department of Statistics, Faculty of Science — University of Colombo, Sri Lanka

GPA: 3.82 / 4.00 **Key Areas:** Machine Learning, Statistical Modeling, Deep Learning, NLP

TECHNICAL SKILLS

Core Languages	Python, SQL, R
ML & DL	Scikit-learn, PyTorch, Feature Engineering, Model Evaluation, Model Quantization (LORA, QLORA)
GenAI / LLM	LLMs, Prompt Engineering, RAG, LangChain, LangGraph, Embeddings, Fine-Tuning (LLaMA 3)
Agentic AI	Multi-Agent Systems, Agentic Memory (Short-term, Semantic, Long-term), Tool Use, Design Patterns
Deployment	FastAPI, Streamlit, Docker, Docker Compose, CI/CD (Jenkins), Vector DBs (Qdrant, Chroma)
Cloud & Data	GCP, Apache Airflow, Terraform, Playwright, BeautifulSoup
AI Interpretability	Grad-CAM, ROUGE-L, LLM-as-a-Judge, Precision@K, Latency Benchmarking

PROJECTS

AI Research Agent | RAG-Based LLM Application

Python · LangChain · OpenAI API · Qdrant · FastAPI · Streamlit · OOP

- Architected an AI-powered research assistant using RAG with LangChain and Qdrant, enabling context-aware question answering over custom document datasets.
- Built a RESTful API backend (FastAPI) for LLM orchestration and developed a Streamlit frontend for real-time semantic querying.
- Applied advanced prompt engineering techniques to reduce hallucination rates and cut inference costs.
- Built a self-improving Reflection Agent pipeline (Draft → Critique → Revise) to automatically refine clarity, accuracy, and completeness of generated responses.
- Developed an evaluation framework using LLM-as-a-Judge scoring across accuracy, completeness, clarity, actionability, and safety metrics; compared ReAct vs Reflection agent performance.
- GitHub: Research-AI-Agent

Kapruka Gift Recommendation Assistant | AI Multi-Agent System

Python · LLM · Multi-Agent Framework · Qdrant Cloud · Memory Systems

- Built an AI gift recommendation engine using RAG and a multi-agent architecture coordinated by an intelligent routing engine.
- Collected and processed 10,000+ product records via Playwright, generated semantic embeddings, and indexed them in Qdrant Cloud for fast retrieval.
- Implemented multi-layer memory (short-term, semantic, long-term) to deliver personalized, context-aware recommendations across sessions.
- GitHub: Gift-AI-Assistant

Prime Land AI Assistant | Advanced RAG System (CAG & CRAG)

Python · Qdrant · OpenAI API · LangChain · Playwright · BeautifulSoup

- Built an end-to-end RAG AI assistant with a custom async data pipeline (Playwright + BeautifulSoup) for structured real estate data ingestion.
- Integrated a Semantic Caching Layer (CAG) and Corrective RAG (CRAG) to significantly improve response reliability and reduce redundant API calls.
- Evaluated system performance using Precision@K, latency profiling, and cost analysis to guide production optimization.
- GitHub: Prime-Land-AI-Assistant

Financial AI Assistant | Fine-Tuned LLM with Hybrid RAG

Python · LLaMA 3 8B · PyTorch · LangChain · LORA · QLORA · Vector DB

- Fine-tuned LLaMA 3 8B Instruct model on a financial domain dataset using parameter-efficient LORA/QLORA techniques.
- Built a Hybrid RAG pipeline combining semantic vector search with document-grounded generation to reduce hallucinations in numerical reasoning.
- Evaluated model rigorously using ROUGE-L, LLM-as-a-Judge scoring, and latency benchmarking; conducted hallucination analysis to drive iterative improvement.
- GitHub: AI-Assistant-For-Financial

Furniture Delivery Prediction System | ML + MLOps Pipeline

Python · Scikit-learn · FastAPI · Docker · Docker Compose · Jenkins · HTML/CSS/JS

- Trained and selected an AdaBoost classifier for predicting furniture delivery outcomes, achieving improved accuracy over baseline models.
- Built a RESTful API (FastAPI) and responsive web interface for model serving; containerized the full application with Docker and Docker Compose.
- Implemented a CI/CD pipeline using Jenkins following MLOps best practices for reproducible, automated deployment.
- GitHub: Furniture-Delivery-Predicting-System